### 3.2.5 Coterminal angles

One complete rotation of a ray in the anticlockwise direction results in an angle measuring of $360°$. By continuing the anticlockwise rotation, angles larger than $360°$ can be produced. If we rotate in clockwise direction, negative angles are produced. Angles $57°$, $417°$ and $-303°$ have the same initial side and terminal side but with different amount of rotations, such angles are called **coterminal angles**. Thus, angles in standard position that have the same terminal sides are coterminal angles . Hence, if $\alpha$ and $\beta$ are coterminal angles, then $\beta = \alpha + k(360°)$, $k$ is an integer. The measurements of coterminal angles differ by an integral multiple of $360°$.

For example, $417°$ and $-303°$ are coterminal because $417° - (-303°) = 720° = 2\,(360°)$.

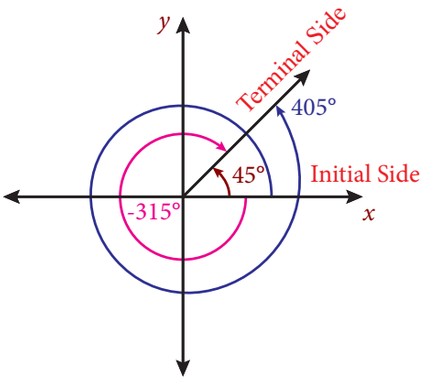



**Figure 3.3**

(i) Observe that $45°$, $-315°$ and $405°$ lie in the first quadrant.
(ii) The following pairs of angles are coterminal angles $(30°, 390°)$; $(280°, 1000°)$ and $(-85°, 275°)$.

### 3.2.6 Basic Trigonometric ratios using a right triangle

We know that six ratios can be formed using the three lengths $a, b, c$ of sides of a right triangle $ABC$. Interestingly, these ratios lead to the definitions of six basic trigonometric functions.

First, let us recall the trigonometric ratios which are defined with reference to a right triangle.

$$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}} \quad ; \quad \cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

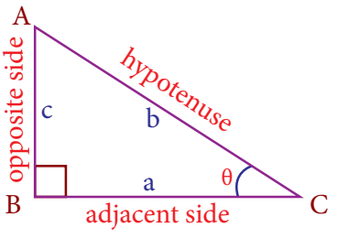



**Figure 3.4**

With the help of $\sin\theta$ and $\cos\theta$, the remaining trigonometric ratios $\tan\theta$, $\cot\theta$, $\text{cosec}\,\theta$ and $\sec\theta$ are determined by using the relations

$$\tan\theta = \frac{\sin\theta}{\cos\theta}\,, \quad \text{cosec}\,\theta = \frac{1}{\sin\theta}\,, \quad \sec\theta = \frac{1}{\cos\theta}\,, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}\,.$$

### 3.2.7 Exact values of trigonometric functions of widely used angles

Let us list out the values of trigonometric functions at known angles.

| $\theta$ | $0°$ | $30°$ | $45°$ | $60°$ | $90°$ |
|---|---|---|---|---|---|
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | $1$ |
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$ | $0$ |
| $\tan\theta$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | undefined |

(i) The values given above are all exact.
(ii) We observe that $\sin 30°$ and $\cos 60°$ are equal. Also $\sin 60°$ and $\cos 30°$ are equal.